

• BUILD WITH AI / CODE FOR COMMUNITY 2026 • TRACK 2: CLEAN AIR & CLIMATE RESILIENCE

“SEE POLLUTION EARLIER. PREDICT ITS MOVEMENT. COORDINATE ACTION.”

AEROLINK BRICS

AI-Powered Federated Air Intelligence & Transboundary Climate Resilience Network
unifying hyper-local ground sensing, orbital remote telemetry, and sovereign cross-
border coordination.

 Sentinel-5P & MODIS Orbital Ingestion

 AeroVision Multimodal Computer Vision

 Lagrangian Transboundary Advection

 Sovereign Federated Learning (FedAvg-Clim)

 Human-in-the-Loop Authority Verification

Author: **Anant
Mishra**

Repository:
github.com/anantmishra7/aerolink-brics

Tech: **React 18 • TypeScript • Vite • Gemini
1.5 Flash**

Why Megacities Miss Hyper-Local & Transboundary Pollution

Current environmental monitoring stations are sparse, reactive, and geographically siloed.

SPATIAL BLINDSPOTS

Hyper-Local Omission

10-20 fixed government stations per megacity create massive 50 km² blindspots. Stubble fires, nocturnal brick kilns, unpaved industrial transit, and fugitive stack bypasses evade municipal sensors completely.

VECTOR INVISIBILITY

Transboundary Transport

Particulate matter (PM2.5) and gaseous precursors (NO2, SO2) do not respect jurisdictional borders. Seasonal winds carry toxic plumes 300+ km under thermal inversion lids, blind-siding downwind megacities.

GEOPOLITICAL FRICTION

Data Siloing Across Borders

Sovereign nations and municipal agencies refuse to upload raw domestic telemetry or industrial emissions to centralized foreign servers, paralyzing coordinated multinational mitigation.

Dimension	Legacy Environmental Monitoring	AEROLINK BRICS Next-Gen Approach
Spatial Resolution	Sparse fixed CAAQMS (10-20 units, ~50 km² gaps)	Fused Grid: Citizen IoT + Mobile Observers + Sub-km Orbital Remote Sensing
Temporal Latency	1 to 3-hour delayed batch telemetry reporting	Real-time stream ingestion with sub-second in-browser AI vision inference
Movement Physics	Static dials showing current AQI without vector dynamics	Lagrangian advection vectors predicting arrival ETAs and plume trajectories
Data Sovereignty	Requires risky centralized data lakes (frequently rejected)	Federated Learning (FedAvg-Clim): Raw data stays local, only encrypted gradients shared

AEROLINK BRICS: Unified Climate Intelligence

Bridging hyper-local ground observations and continental atmospheric dynamics.

4.5 - 6.0 hrs

EARLIER TRANSBOUNDARY WARNING

4 Sources

ATTENTION-WEIGHTED DATA FUSION

+7.4 %

FEDERATED ACCURACY GAIN (E=0.75)

100 %

HUMAN-IN-THE-LOOP ACCOUNTABILITY

CORE PILLAR 01 & 02

Multi-Source Fusion & Optical AI

- ✓ **Dynamic Attention-Weighted Data Fusion:** Dynamically balances Satellite (30%), Ground (25%), Meteorology (25%), and Citizen reports (20%).
- ✓ **AeroVision Multimodal Vision Service:** Leverages Google Gemini 1.5 Flash to classify smoke, industrial flaring, and agricultural burning from citizen photos with optical confidence.
- ✓ **Orbital Remote Sensing Correlation:** Real-time cross-validation against Copernicus Sentinel-5P (NO2/SO2) and MODIS/VIIRS thermal fire radiance (MW/km²).

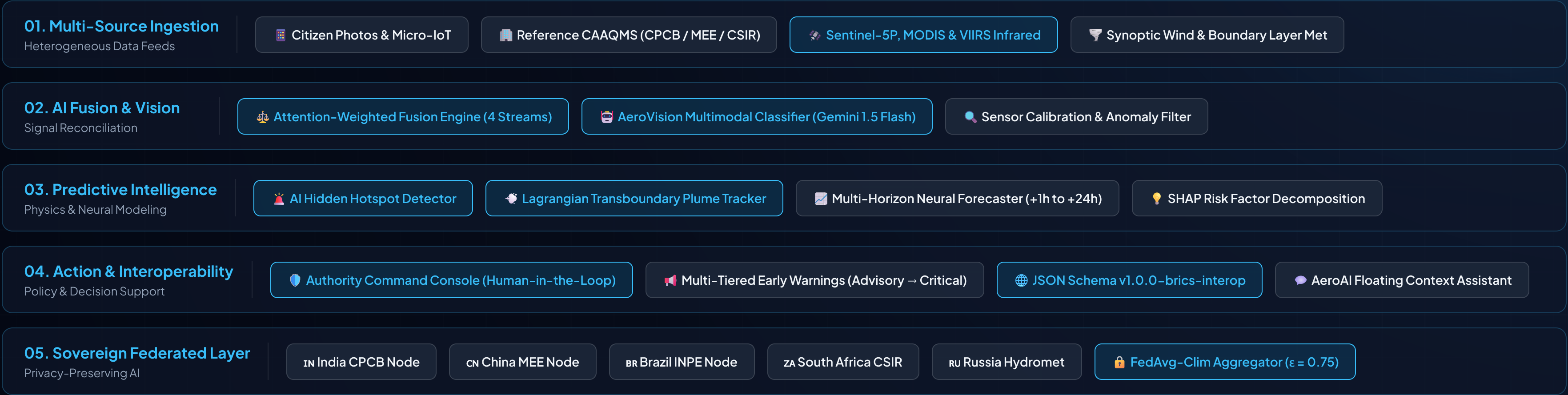
CORE PILLAR 03 & 04

Predictive Physics & Sovereign Coordination

- ✓ **Lagrangian Plume Advection Tracker:** Calculates real-time wind vectors, atmospheric boundary layer heights, and downwind arrival ETAs across national borders.
- ✓ **Economic Corridor Risk Monitoring:** Live monitoring of critical supply chains (e.g. Delhi NCR-Agra, Beijing-Tianjin-Hebei, Mpumalanga Coal Belt).
- ✓ **Sovereign Federated Learning (FedAvg-Clim):** Trains collaborative dispersion models across BRICS nodes with Differential Privacy without exporting domestic sensor logs.

Full-Stack Five-Tier Modular Architecture

Resilient data pipelines from edge IoT ingestion to international policy execution.



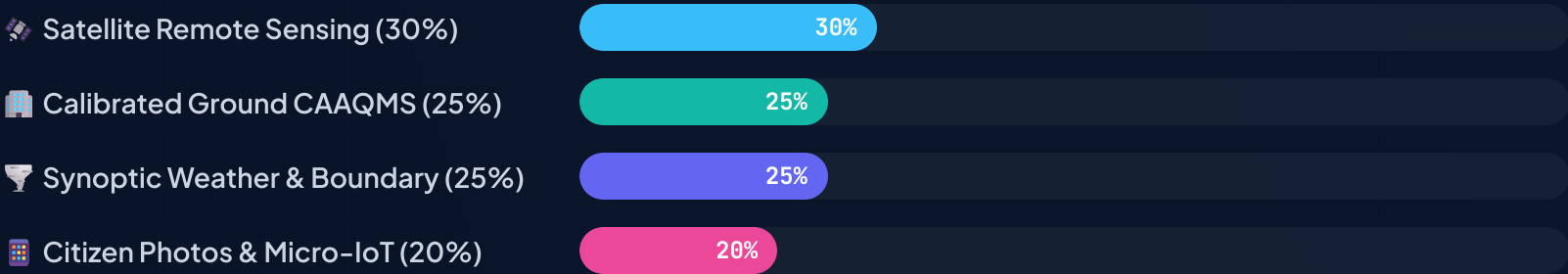
Data Fusion Engine & AeroVision Optical AI

Transforming raw citizen uploads and multi-spectral satellite telemetry into actionable signals.

DYNAMIC ATTENTION FUSION

Balancing Disparate Quality Streams

Raw citizen inputs or single satellite passes cannot be trusted in isolation. The engine applies dynamic attention weighting across 4 independent environmental channels:

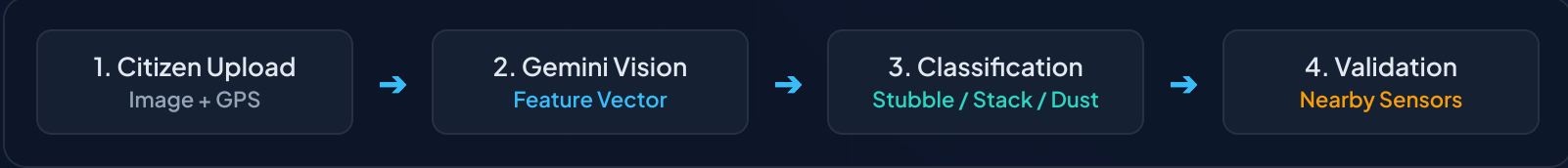


Adaptive Drift Compensation: If cloud cover blinds Sentinel-5P, weights instantly reallocate toward ground sensors and meteorological advection vectors.

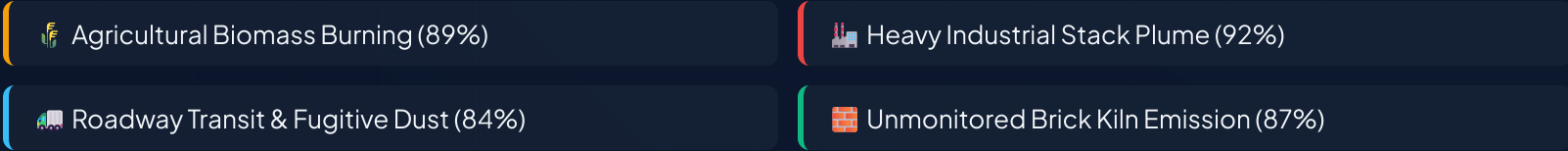
GEMINI 1.5 FLASH VISION

AeroVision Optical Analysis Pipeline

Citizens submit geolocated smartphone photos of pollution plumes. The model extracts optical characteristics (plume opacity, color hue, dispersion geometry) to predict source etiology:



Classification Confidence Categories:



Transboundary Plume Tracking & Corridor Risk

Tracking cross-border advection vectors and arterial trade routes across BRICS economies.

ADVECTION PHYSICS

Lagrangian Trajectory Modeling

Pollution does not stay stationary. AeroLink calculates forward trajectory vectors by coupling synoptic wind fields with nocturnal planetary boundary layer heights:

Simulated Plume Vector: #TRK-2026-IND-01

ACTIVE ADVECTION

- Origin: Punjab-Haryana Agricultural Margin (30.3° N, 75.8° E)
- Destination: Delhi NCR Urban Basin (28.6° N, 77.2° E)
- Transit Distance: 320 km | Advection Speed: 24 km/h NW → SE
- Inversion Ceiling: 45 meters (Critical Trapping Layer)
- Calculated Arrival ETA: 4.5 - 6.0 Hours

Provides authorities with decisive advance warning before high-density PM2.5 can descend upon urban populations.

ECONOMIC CORRIDORS

Arterial Freight & Megacity Belts

Monitors cumulative risk across vital manufacturing hubs where emissions overlap with heavy logistics corridors:

IN Delhi NCR → Agra Industrial Corridor
Status: CRITICAL (AQI 378 Peak Expected)

+44% Risk

CN Jing-Jin-Ji Megacity Region (Beijing-Tianjin-Hebei)
Status: HIGH WATCH (NO2 Thermal Stagnation)

+28% Risk

ZA Mpumalanga Coal Belt → Gauteng Hub
Status: ELEVATED (SO2 Plume Drift)

+22% Risk

BR Santos Maritime Port → São Paulo Metropole
Status: MODERATE (Maritime Sea Breeze Dispersal)

Nominal

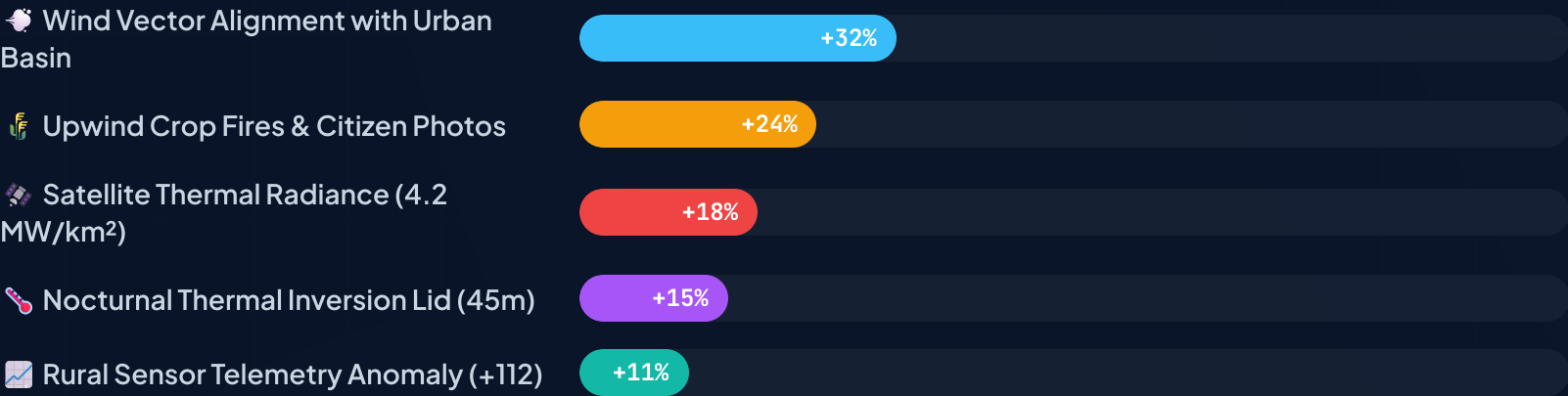
Explainable AI (XAI) & Multi-Horizon Forecasting

Eliminating black-box mistrust by decomposing causal drivers with probabilistic bounds.

SHAP-DECOMPOSED ATTRIBUTION

"Why Is Risk Increasing Right Now?"

Decision makers will not enforce economic restrictions or dispatch emergency crews based on opaque neural network outputs. AeroLink breaks down composite risk scores into quantifiable factors:



Plain-Language Synthesis: Sustained 24 km/h north-westerly winds are advecting biomass smoke under a 45m inversion cap directly into the Delhi metropolitan basin.

PROBABILISTIC FORECASTER

Multi-Horizon Predictions with 90% CI

Predicts future AQI and PM2.5 across +1h, +3h, +6h, +12h, and +24h windows with shaded uncertainty bounds reflecting meteorological turbulence:

Horizon	Predicted AQI	PM2.5 (µg/m³)	Confidence Interval (90%)
Current (0h)	214 (Poor)	112.4	208 – 220
+3 Hours	285 (Very Poor)	168.0	260 – 310
+6 Hours (Peak)	378 (Severe)	236.8	340 – 415
+12 Hours	295 (Very Poor)	175.2	250 – 340
+24 Hours	228 (Poor)	121.0	180 – 275

Forecast Engine: GRU-Transformer Hybrid

Peak Window: 18:00 – 22:00 IST

Authority Response Center: Human-in-the-Loop

AI Recommends → Human Verifies → Authority Acts with full legal and regulatory chain-of-custody.

PROTOCOL 1

1. Anomaly Flagged

AI triggers early warning alert #ALT-IN-2026-004. Anomaly score 0.94 correlates satellite fire pixels with rural IoT surge.

PROTOCOL 2

2. Human Verification

Environmental Duty Officer inspects optical evidence, wind vectors, and model attributions. Zero automated fines or closures.

PROTOCOL 3

3. Coordinated Dispatch

Officer triggers verified action: Anti-smog mist cannon deployment, inter-state advisory dispatch, and Stage-IV GRAP enforcement.

AUDITABLE ACTION MATRIX

Six Standard Operational Mitigations

 Dispatch Mobile Rapid Unit

Deploys smog cannons and mobile inspection vans directly to upwind hotspots.

 Issue Public Health Advisory

Sends targeted mobile broadcast alerts to schools, hospitals, and high-risk citizens.

 Notify Neighboring Region

Transmits structured transboundary alert payload via JSON schema to adjacent state/country.

 Industrial Throttling / GRAP

Temporarily dials back coal thermal outputs and pauses non-essential road construction.

 Ramp IoT Telemetry Frequency

Commands nearby sensor mesh to increase sampling frequency from 15 min to 1 min.

 Post-Mitigation Resolution Log

Logs full forensic chain for historical model retraining and compliance audits.

Federated Climate AI: FedAvg–Clim Protocol

Collaborative multinational intelligence with zero raw data leaves sovereign borders.

SOVEREIGN NODE CLUSTER

Participating National Nodes

Each nation hosts an autonomous sovereign edge container deployed inside its domestic infrastructure:

IN India (CPCB Cloud, New Delhi)	Active • Round 18
CN China (MEE Supercomputing, Beijing)	Active • Round 18
BR Brazil (INPE Earth Observation, São Paulo)	Active • Round 18
ZA South Africa (CSIR Meraka, Pretoria)	Active • Round 18
RU Russia (Roshydromet Hydro–Met, Moscow)	Active • Round 18

Differential Privacy Guarantee: Gaussian noise injected with privacy budget $\epsilon = 0.75$, $\delta = 10^{-5}$, guaranteeing that individual industrial plants or telemetry points cannot be reverse-engineered.

GLOBAL MODEL CONVERGENCE

Federated Aggregation Metrics

Iterative rounds of FedAvg–Clim combine localized atmospheric insights into a robust global neural model:

Round	Nodes	Loss (MSE)	Accuracy Gain	Status
Round 01	5 / 5	0.482	Baseline	Converged
Round 06	5 / 5	0.312	+3.2%	Converged
Round 12	5 / 5	0.219	+5.8%	Converged
Round 18	5 / 5	0.164	+7.4%	Current Active

Why It Matters to International Law:

Solves the fundamental political dilemma of transboundary climate agreements: Nations gain superior predictive power from neighboring meteorological phenomena without surrendering sensitive sovereignty data.

Judge Demo Mode: Cross–Border Smoke Event

A ten-step end-to-end evidence chain demonstrating complete operational maturity.



Technology Stack & Scientific Standards

Modern web architecture designed for low-latency command centers and high-fidelity rendering.

FULL-STACK IMPLEMENTATION

Production-Grade Architecture

- ✓ **Frontend Framework:** React 18, TypeScript, Vite 5 for sub-millisecond hot reloading and type-safe state governance.
- ✓ **Geospatial Vector Engine:** Leaflet & Custom Animated SVG Vector Mesh rendering 12 interactive layers with zero frame drops.
- ✓ **Data Visualization:** Recharts with confidence interval bands and historical federated training curves.
- ✓ **Multimodal AI:** Google Gemini 1.5 Flash Vision API integration paired with local client-side fallbacks.
- ✓ **Open Interoperability:** JSON Schema v1.0.0-brics-interop for REST & gRPC enterprise integration.

SCIENTIFIC INTEGRITY

Ethics, Transparency & Standards

- ✓ **Explicit Simulation Badges:** Every simulated or demo dataset displays prominent [SIMULATION / DEMO DATA] indicators to prevent misrepresentation.
- ✓ **Prudent Optical Classification:** Models never claim definitive legal causality from images alone; labels are classified as *"Potential Source Estimates"* requiring multi-sensor cross-validation.
- ✓ **Zero Algorithmic Punishment:** Algorithmic suggestions cannot automatically issue citations, fine factories, or halt transit without human officer approval.
- ✓ **Data Locality Guarantees:** Compliant with national data sovereignty laws (India DPDP Act, China PIPL, Brazil LGPD).

Impact, Strategic Roadmap & Conclusion

Transforming international air quality governance from reactive despair to proactive resilience.

QUANTIFIABLE IMPACT

4.5 – 6.0 Hours

Advance warning window enables hospital emergency wings to prepare respiratory wards, schools to cancel outdoor sports, and dust-suppression cannons to deploy before plumes hit.

ECONOMIC SAVINGS

Logistics Resilience

Rerouting freight corridors prior to acute smog lock-ins saves millions in perishable goods and prevents catastrophic multi-vehicle highway pile-ups in dense smog.

DIPLOMATIC BREAKTHROUGH

Sovereign Commons

Unlocks mutual climate collaboration between BRICS allies without geopolitical deadlock or fear of intelligence vulnerability.

DEVELOPMENT ROADMAP

Next-Phase Milestones

- **Phase 1 (Q4 2026):** On-device WebAssembly/TensorFlow.js edge model deployment for offline citizen plume detection.
- **Phase 2 (Q1 2027):** Direct orbital tasking integration with the BRICS Earth Observation Satellite Constellation.
- **Phase 3 (Q2 2027):** Expansion to maritime shipping emissions corridors in the Indian and Atlantic Oceans.

PROJECT LINKS & ACCESS

Experience the Platform

- **GitHub Repository:** github.com/anantmishra7/aerolink-brics
- **Local Development:** `npm run dev` on `http://localhost:5173`
- **Judge Demo:** One-click interactive walkthrough in platform header
- **Contact:** Anant Mishra (anantm408@gmail.com)